

The background is a high-contrast, abstract composition. It features a light gray, textured surface, possibly resembling watercolor or paper. Overlaid on this are various black ink elements: large, irregular splatters in the corners; fine, radiating lines and dots; and some darker, more solid black shapes. The overall effect is one of dynamic, organic movement.

project PRESENTATION

TASK 3: DATA ANALYSIS WITH PANDAS

TITLE:

STUDENT MARKS DATA ANALYSIS
USING PYTHON PANDAS

- **Objectives:**
- **The objective of this project is to perform basic data analysis using the Pandas library. The project demonstrates how to load a CSV file, inspect the data, clean missing values, perform filtering, grouping, and calculate summary statistics.**

TOOLS USED:

PYTHON 3

PYDROID 3

PANDAS LIBRARY

CSV FILE


```

1 import pandas as pd
2
3 # Student data
4 data = {
5     "Name": ["Ammu", "Rahul", "Priya", "Kiran", "Sita"],
6     "Marks": [95, 80, None, 75, 90]
7 }
8
9 # Create DataFrame
10 df = pd.DataFrame(data)
11
12 print("Original Data")
13 print(df)
14
15 # Fill missing values
16 df["Marks"] = df["Marks"].fillna(df["Marks"].mean())
17
18 print("\nData after cleaning")
19 print(df)
20
21 # Students scoring above 80
22 print("\nStudents scoring above 80")
23 print(df[df["Marks"] > 80])
24
25 # Average marks
26 print("\nAverage Marks:")
27 print(df["Marks"].mean())

```

IMPORT PANDAS AS PD

CREATE SAMPLE STUDENT DATA

```

DATA = {
    "NAME": ["AMMU", "RAHUL", "SNEHA", "KIRAN",
             "PRIYA"],
    "MARKS": [95, 82, 76, 90, 88],
    "DEPARTMENT": ["CSE", "CSE", "ECE", "ECE", "CSE"]
}

```

CREATE DATAFRAME

DF = PD.DATAFRAME(DATA)

PRINT("STUDENT DATA")

PRINT(DF)

PRINT("\NAVERAGE MARKS:")

PRINT(DF["MARKS"].MEAN())

PRINT("\NSTUDENTS WITH MARKS ABOVE 85:")

PRINT(DF[DF["MARKS"] > 85])

PRINT("\NDEPARTMENT-WISE AVERAGE MARKS:")

PRINT(DF.GROUPBY("DEPARTMENT")

["MARKS"].MEAN())

Code output:

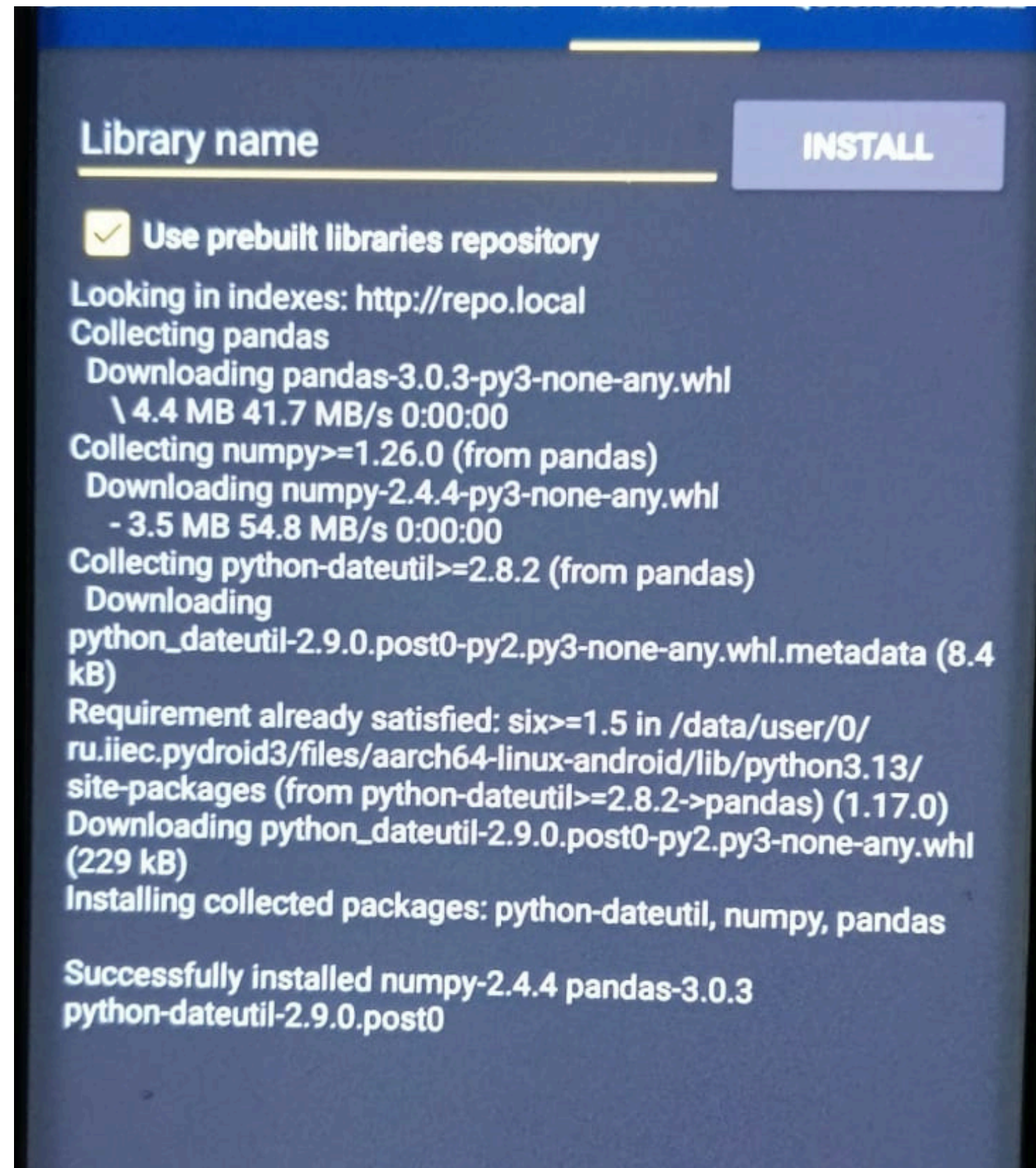
```
Original Data
  Name Marks
0  Ammu  95.0
1  Rahul 80.0
2  Priya  NaN
3  Kiran 75.0
4  Sita  90.0

Data after cleaning
  Name Marks
0  Ammu  95.0
1  Rahul 80.0
2  Priya 85.0
3  Kiran 75.0
4  Sita  90.0

Students scoring above 80
  Name Marks
0  Ammu  95.0
2  Priya 85.0
4  Sita  90.0

Average Marks:
85.0

[Program finished]
```



EXPLANATION:

THIS PROJECT USES THE PANDAS LIBRARY TO ANALYZE STUDENT MARKS. A DATAFRAME IS CREATED WITH STUDENT NAMES, MARKS, AND DEPARTMENTS. THE PROGRAM CALCULATES THE AVERAGE MARKS, FILTERS STUDENTS WHO SCORED ABOVE 85, AND GROUPS THE DATA BY DEPARTMENT TO CALCULATE THE AVERAGE MARKS OF EACH DEPARTMENT.

INSIGHTS:

THE OVERALL AVERAGE MARKS ARE
86.2.

THREE STUDENTS SCORED MORE
THAN 85 MARKS.

THE CSE DEPARTMENT HAS A HIGHER
AVERAGE SCORE THAN THE ECE
DEPARTMENT.

CONCLUSION:

THIS PROJECT HELPED ME UNDERSTAND HOW TO USE THE PANDAS LIBRARY FOR DATA ANALYSIS. I LEARNED HOW TO CREATE A DATAFRAME, CALCULATE AVERAGES, FILTER DATA, GROUP RECORDS, AND ANALYZE INFORMATION IN A SIMPLE AND EFFECTIVE WAY.

THANK

you